

CEMENT QUALITY CONTROL SYSTEM



25-26J-250

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AGENDA

Introduction to Problem

01

Connection Between Components

02

Design Excellence

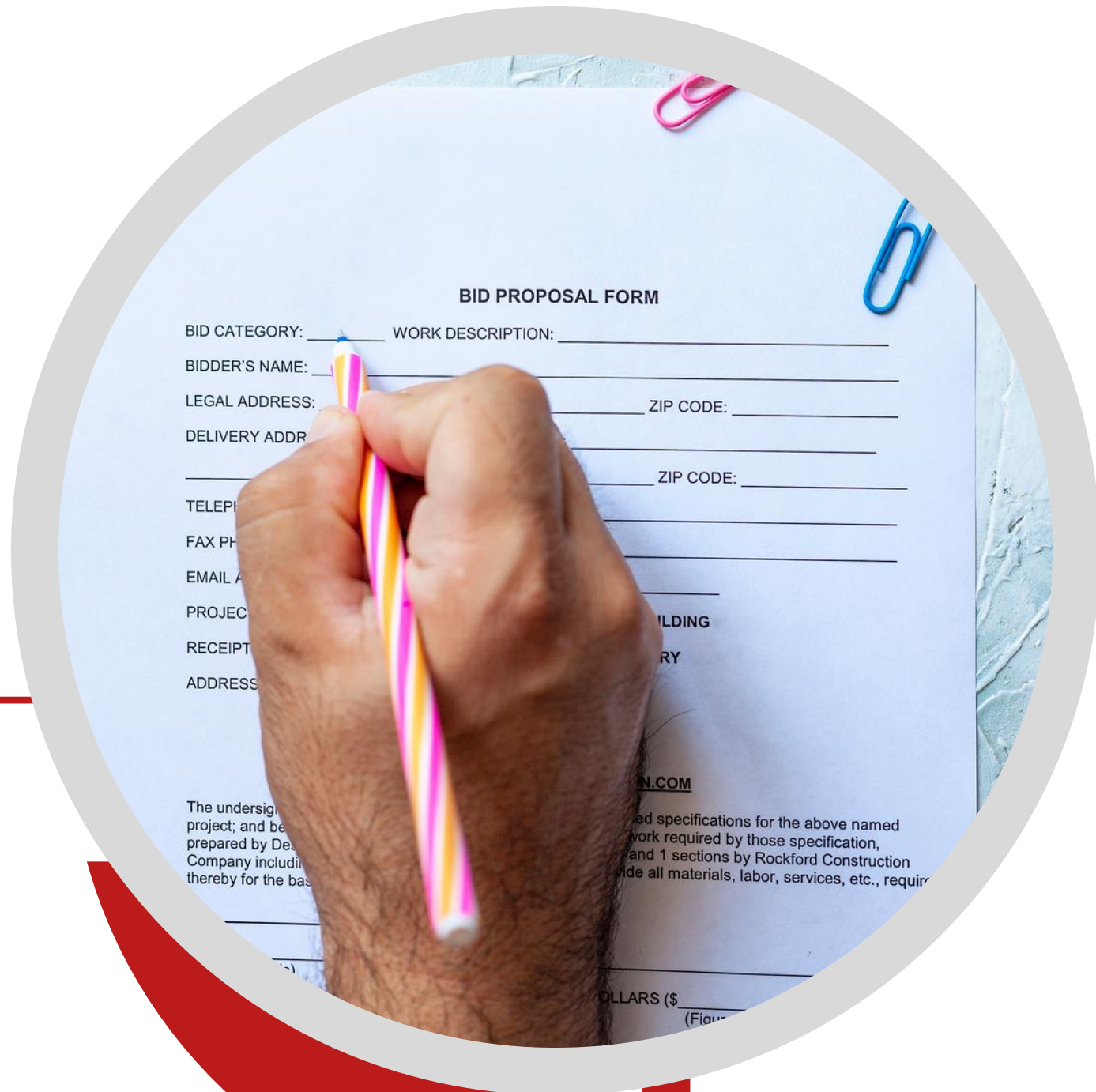
03

Four Components

04

Commercialization

05



OUR TEAM



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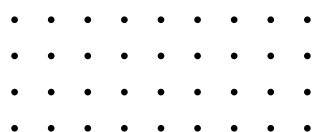


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INTRODUCTION TO PROBLEM



- ❖ Manual QC methods → slow (15–60 mins) & expert-dependent
- ❖ Cube testing → Manual measurement, error-prone
- ❖ Environmental factors (temp, humidity) ignored
- ❖ Delays cause cost, inconsistency, weak QC



CEMENT QUALITY CONTROL SYSTEM

DESIGN EXCELLENCE



System Architecture Excellence

- Modular design with 4 integrated components
- Technology-agnostic approach (TensorFlow, PyTorch, Scikit-learn)
- Microservices pattern with unified API gateway



Technical Excellence

- Full-stack integration (React + FastAPI)
- Efficient model loading (~5s total, <400ms inference)
- Robust testing & CI/CD ready
- Production-grade security & error handling



Research Contributions

- 99%+ time reduction across all QC processes
- First Sri Lankan OPC-specific AI platform
- Novel hybrid modeling & feature engineering

AI-POWERED CEMENT CLINKER PHASE CLASSIFICATION: ENHANCING QUALITY CONTROL IN HF-ETCHED MICROSCOPY IMAGE ANALYSIS



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CEMENT CLINKER PHASE CLASSIFICATION

PROBLEM

Manual petrographic analysis of clinker phases is **slow (1–2 hrs sample)** and depends on **expert skills**.

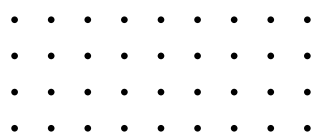
USER REQUIREMENT AND SOLUTION

- **Quick clinker phase identification**
→ AI automatically classifies clinker images.
- **Reduce manual analysis errors**
→ Deep learning ensures consistent results.
- **Easy image upload & usage**
→ Simple web-based interface.
- **Fast quality control decisions**
→ Real-time predictions with confidence scores.



CEMENT CLINKER PHASE CLASSIFICATION

- 01 Data:** 300 cement clinker microscopic images from **INSEE Cement**
- 02 Model:** **MobileNetV2 (Keras + TensorFlow)** for clinker phase classification
- 03 System:** **FastAPI + React** for real-time image upload and prediction
- 04 Contribution:** Automated, web-based clinker phase analysis system



Automate Color-Based Particle Analysis of Cement Raw Meals Microscopic Images Using Machine Learning

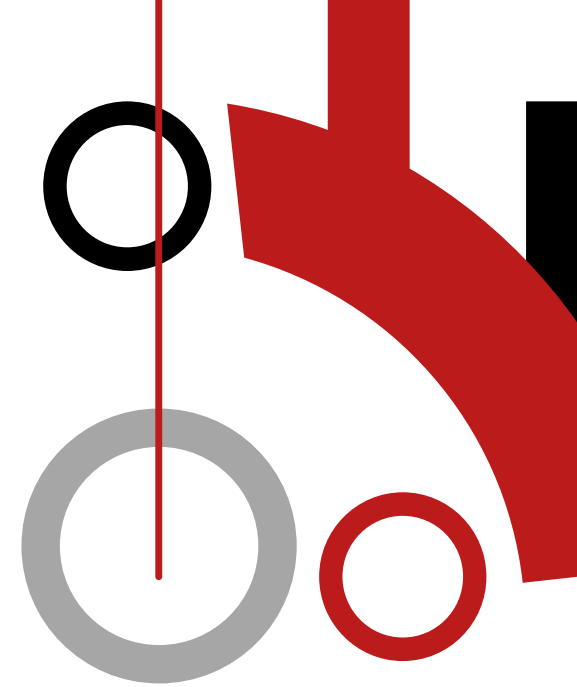
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Automate Color-Based Particle Analysis of Cement Raw Meals Microscopic Images Using Machine Learning



Problem

- Cement raw meal quality analysis is traditionally performed manually by experts using microscopic images.
- Manual particle identification based on color and visual inspection is :
 - Time consuming.
 - Prone to human error.

Solution

- Develop ML model to automatically detect cement raw meals particles and classify them based on color. (dark red, light red, white)
- Using the developed ML model and count the particles accurately.

Automate Color-Based Particle Analysis of Cement RawMeals Microscopic Images Using Machine Learning

User Requirements

- Ability to upload microscopic images of cement raw meals.
- Automatic detection and classification of particles by color.
- Accurate counting of individual particles.
- Fast analysis to support real-time decision making.
- Ability to review previous predictions.

User Feedback on Prototype

- Users found the system easy to use and intuitive.
- Automated analysis significantly reduced the time required for particle counting.

Automate Color-Based Particle Analysis of Cement RawMeals Microscopic Images Using Machine Learning

Design Excellence / Contribution

- **Data Collection:** 300 microscopic images from INSEE company.
- **Model Training:**
 1. YOLO V9: Performs particle object detection, classification and counting of microscopic images.
- **FastAPI (Python):** RESTful API for image upload and real-time predictions.
- **React:** Interactive and user-friendly frontend and Image upload, result visualization, and prediction history.
- **Research Contribution:** Provides a real-time, web-based solution for raw meal particle analysis

IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube Compressive Strength Testing

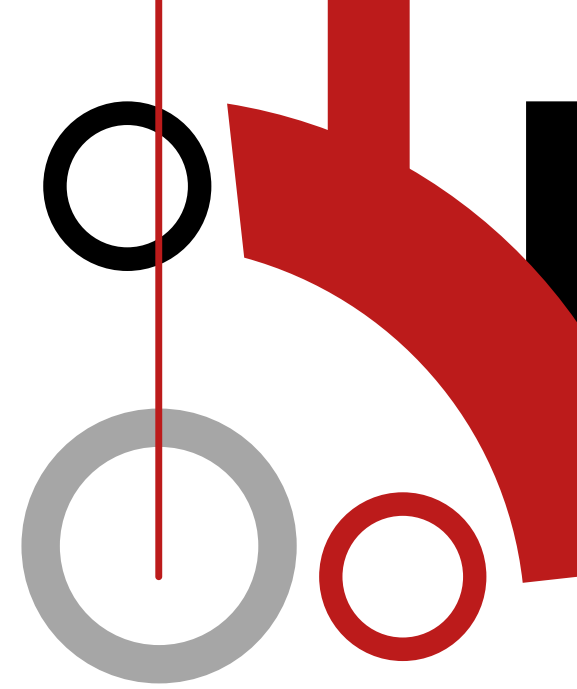


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IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube Compressive Strength Testing



Problem Statement



- Why is IoT-Based Area Measurement and Image-Based Crack Detection Important for Cement Companies like INSEE?

Solution



- An IoT-based system measures cement cube area automatically using distance sensors, while a camera with a CNN model detects cracks in real time during compression testing. This reduces manual work, improves safety, minimizes errors, and enables automatic compressive strength calculation with digital record keeping.

IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube Compressive Strength Testing

Design Excellence / Contribution

- **Data Collection:** Cement cube surface images during compression testing and IoT-based distance sensor data.
- **Model Training:**
 1. U-Net
 2. U-Net++
 3. DeepLabV3+
- **Model Selection:** U-Net segmentation model achieved the best crack detection accuracy and is selected for real-time implementation.
- **System Integration:** IoT sensors automatically calculate cube area, while a web dashboard displays real-time measurements, crack alerts, and test history.
- **Research Contribution:** A novel IoT and image-based automated cement cube testing system that reduces manual effort, improves safety, and enables digital reporting.

IoT-Based Area Measurement and Image-Based Crack Detection for Cement Cube Compressive Strength Testing

User Requirements

- User Profile: Sri Lankan cement laboratory technicians, quality control engineers, and plant supervisors.
- Automated cube area measurement without manual tools.
- Real-time crack detection during compression testing.
- Immediate compressive strength calculation after crack detection.
- Accurate results suitable for Sri Lankan cement testing standards.

User Feedback on Prototype

- Real-time visual crack alerts make it easy to identify failure instantly, even in busy lab environments.
- Automated measurements closely match manual lab results within acceptable tolerance levels.
- Significantly reduces testing time and minimizes operator involvement.

A HYBRID MODELLING FRAMEWORK FOR EARLY AND LONG-TERM CEMENT STRENGTH PREDICTION

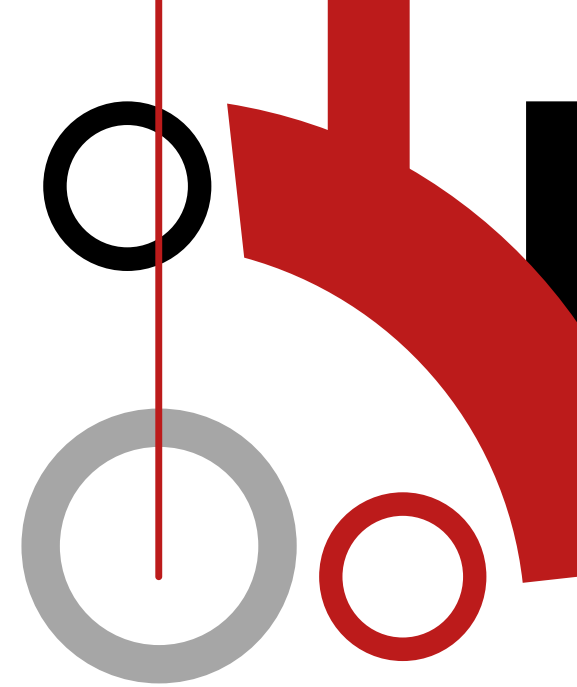
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A HYBRID MODELLING FRAMEWORK FOR EARLY AND LONG-TERM CEMENT STRENGTH PREDICTION



Problem Statement



- Existing studies use foreign datasets, specialty cements, or post-production data – not suited for Sri Lankan OPC real-time conditions.
- Environmental factors (temperature, humidity) and process data are often overlooked.
- Traditional lab testing requires 1–56 days of waiting time, delaying quality control decisions.

Solution



- Hybrid Multi-Output Ensemble Model combining XGBoost and LightGBM algorithms.
- Provides real-time predictions for both early-age and long-term strength at 5 time intervals (1D, 2D, 7D, 28D, 56D).
- Integrates plant process data + chemical composition data

A HYBRID MODELLING FRAMEWORK FOR EARLY AND LONG-TERM CEMENT STRENGTH PREDICTION

Design Excellence / Contribution

- **Data Collection:** Sri Lankan OPC plant data + lab test results
- **Feature Engineering:** 30+ derived features integrating process & chemical data.
- **Dual Model Training:**
 1. XGBoost: Handles non-linear interactions and complex patterns.
 2. LightGBM: Efficient gradient boosting with faster training.
- **FastAPI (Python):** RESTful API for real-time predictions.
- **React:** Interactive UI for cement plant operators and QC managers.
- **Research Contribution:** First hybrid model specifically for Sri Lankan OPC with real-time capability.

A HYBRID MODELLING FRAMEWORK FOR EARLY AND LONG-TERM CEMENT STRENGTH PREDICTION

User Requirements

- User Profile: Sri Lankan cement plant operators, quality control managers, and production supervisors.
- Early and long-term strength prediction in real-time
- Real-time input of cement composition and plant process parameters.
- Proper validation for Sri Lankan OPC typical ranges
- Support for decimal precision required in INSEE.

User Feedback on Prototype

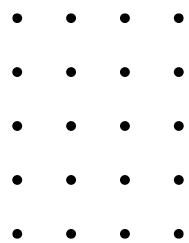
- Visual indicators make it easy to understand results at a glance even on busy production days.
- Predictions closely match our lab results within acceptable margins for local cement.
- Reduces waiting time from 56 days to seconds.

CEMENT QUALITY CONTROL SYSTEM

COMMERCIALIZATION



- ❑ **Target Audience:** INSEE QC engineers and lab staff in cement plants.
- ❑ **Uniqueness:**
 1. AI-based web tool for fast, accurate clinker analysis.
 2. 4-in-1 platform vs single-purpose competitors
- ❑ **Market Size:** Growing demand across the global cement industry for automated QC. (*Our target: Rs. 400-600M in QC automation*)



THANK YOU!

